EEG-Based Mental Health Studies — (Wiley Publisher)

Stress Detection | Depression Detection | Multimodal Approaches | Survey & Gap Papers

S.No	Year	Authors	Dataset	Modality	Methodology / Model Used	Accuracy	Identified Gaps	Category
1	2022	Hatem S. A. Hamatta et al.	DEAP Dataset	EEG + PPG + GSR	Genetic Algorithm + ECNN-LSTM	99.9%	Limited comparison with advanced feature extraction approaches; future enhancement required for improving classification robustness and feature optimization.	EEG Stress
2	2025	Deep Learning Mental Stress Prediction paper authors	EEG Psychiatric Disorders Dataset (Kaggle)	EEG	AFAWT + STFT-RT + YDSE + ParNeXt v1	Optimized (exact % not reported)	EEG noise sensitivity, subject variability, environmental interference, classifier overfitting, and large labeled-data requirements reduce robustness and real-time applicability.	EEG Stress
3	2024	Md Raihan Khan, Mohiuddin Ahmad	GAMEEMO + SEED	EEG	Random Forest + RNN + PSD Features	RNN: 87% (Arousal), 83% (Valence)	EEG non-stationarity, noise, feature extraction complexity, and limited research specifically targeting academic mental stress remain challenges.	EEG Stress
4	2020	R. Gupta, M. Alam, P. Agarwal	EEG Stress Dataset	EEG	Modified SVM + Whale Optimization	Not explicitly reported	Scalability limitations and reduced generalization capability across diverse EEG stress scenarios.	EEG Stress
5	2024	Comparative ML Mental Stress Detection Paper	EEG Dataset	EEG	KNN + SVM + RF + EDA analysis	KNN: lowest MAE (0.62)	Small and noisy datasets limit deep learning adoption; model robustness can improve through stronger preprocessing and larger datasets.	EEG Stress
6	2023	Nazmus Sakib, Md Kaful Islam, Tasnuva Faruk	Custom EEG Dataset (32 subjects), PHQ-9	EEG	KNN + SVM + Hjorth Parameters + Shannon Entropy + Log Energy Entropy	98.43%	Small participant pool (32 subjects) and limited demographic diversity constrain broader clinical generalization.	EEG Depression
7	2024	AI Analysis EEG Depression paper authors	Clinical EEG + Evoked Potential observations	EEG + Evoked Potential	AI-assisted EEG analysis	Diagnostic support (no % reported)	EEG biomarkers alone may not provide sufficient diagnostic reliability; broader multimodal clinical evaluation is needed.	EEG Depression
8	2025	EEG ML/DL Depression Survey authors	Multiple EEG Depression papers	EEG	CNN + RNN + Hybrid DL + Survey Analysis	N/A (Survey Paper)	Heavy reliance on binary classification and insufficient severity-level prediction; multimodal integration remains underexplored.	EEG Depression

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9	2026	Weitong Guo et al.	AVEC2014 + Chinese localized depression dataset	Audio + Video	Transformer Cross Modal Coattention Network (CMCAN)	83.9%	Cross-modal temporal synchronization and preservation of modality-specific information remain difficult in multimodal fusion.	Multimodal
10	2025	IIOT Stress Assessment paper authors	Proprietary multimodal sensing framework	EEG + HRV + GSR + Voice + Behavioral + Environment	Transformer Attention + Federated Learning + Multimodal Fusion	+3.6% improvement over TFN	Privacy constraints, multimodal synchronization complexity, noisy sensor conditions, and real-world deployment challenges affect robustness.	Multimodal
11	2025	Hakan Uyanik et al.	74 reviewed studies	EEG	ML + DL Systematic Review	N/A (Review Paper)	Larger EEG datasets, improved model interpretability, and stronger generalization across populations are needed.	Survey / Gap
12	2025	EEG-MDD Survey paper authors	31 EEG-MDD papers	EEG	CNN + RNN + LSTM + Hybrid DL Survey	N/A (Survey Paper)	Small homogeneous datasets and limited demographic diversity reduce model generalization capability.	Survey / Gap
13	2025	EEG ML/DL Depression Survey authors	Multiple EEG studies	EEG	Survey + Gap Analysis	N/A (Survey Paper)	Limited multimodal fusion strategies and inadequate depression severity-stage prediction remain open research directions.	Survey / Gap

Color Legend: Blue = EEG Stress Yellow = EEG Depression Green = Multimodal Orange = Survey / Gap Papers